

Oakwood AI Innovation Lab

Final strategic package for Dr. Shoshana Smith and Oakwood leadership

[Package](#)[Thesis](#)[Budgets](#)[Architecture](#)[Roadmap](#)

FINAL BRIEFING BUNDLE · MAY 2026

Build the lab where Oakwood students learn AI by shipping.

A dark, presentation-ready strategic package fusing the strongest web, PowerPoint, and research directions into one coherent story: student outcomes, local AI capability, responsible governance, and a practical path from funding window to demo day.

[Download PDF](#)[Download PowerPoint](#)[Download Full Bundle](#)

One main experience. Three supporting artifacts.

The website is the strongest room-facing experience. The PDF and PowerPoint are attached for administrators who need a file, a printout, or a deck they can circulate internally.

01

Immersive HTML Briefing

This page is the recommended primary link. It carries the strongest UI/UX, dark theme, visuals, and integrated narrative.

[Current page](#)

02

PDF Briefing

Best for emailing, printing, or attaching as a static administrative artifact.

[Open PDF](#)

03

PowerPoint Deck

Best if Dr. Smith or the department wants to edit, reorder, or reuse slides for an internal meeting.

[Open PPTX](#)

Do not build a computer lab. Build a production studio.

The strongest version of this proposal is not hardware-first. It is capability-first: students using AI, building AI systems, running local models, and governing AI responsibly.

PILLAR 01

Use AI

Prompting, research, design, coding, analysis, presentations, and multimodal workflows with academic integrity rules.

PILLAR 02

Build AI

Apps, agents, retrieval systems, dashboards, automations, voice, vision, and campus-facing prototypes.

PILLAR 03

Run Local

Ollama, vLLM, quantization, GPU constraints, model serving, privacy, latency, and cloud-to-edge tradeoffs.

PILLAR 04

Govern AI

Data classification, human approvals, audit trails, prompt injection tests, bias checks, and responsible deployment.



The room should signal AI production the moment someone walks in.

Ten seats matter, but the zones matter more: build stations, local models, demo wall, studio corner, security table, and edge AI bench.

Oakwood has a distinct voice in AI.

HBCU, Adventist, STEM, and Huntsville proximity form a credible strategic lane: technical skill joined to service, stewardship, human dignity, and workforce readiness.

HBCU workforce relevance

The lab can connect students to one of the fastest-moving economic shifts in the country while producing proof of capability.

Adventist ethics angle

Oakwood can speak to AI through human dignity, stewardship of data, service, restraint, and responsible use.

Huntsville pipeline

Defense, aerospace, cyber, engineering, health, and local startups create natural demo-day and internship pathways.

Four budget paths. One strategic ladder.

Each tier should be presented as a complete outcome path, not a raw shopping list. The choice depends on the funding window, procurement deadline, and desired institutional signal.

Starter Pod

\$20K

Pilot path for fast proof of student interest.

- 4-6 capable student seats
- Shared local AI or edge AI node
- Display, basic storage, open-source stack
- Best for seed funding

Production Lab

\$50K

Recommended minimum for a credible launch.

- 6-10 strong seats
- DGX Spark-class local node
- Demo wall, NAS, managed network
- 8-week pilot curriculum

Research Lab

\$100K

Stronger institutional signal for faculty and grants.

- Mixed premium seats
- Local AI node plus GPU workstation
- Studio and edge bench
- Employer MOU and research framing

Flagship Center

\$200K

Donor-ready center with visibility and naming potential.

- 10 premium seats
- Multiple local AI nodes
- GPU server and full AV environment
- External partner recruitment

Recommendation: lead with the \$50K tier if the funding window is uncertain. Move to \$100K if Dr. Smith has a real grant or donor lane. Save \$200K for donor/naming or broader institutional capital.

Mixed architecture teaches the whole field.

Students need daily-driver machines, local model systems, shared GPU horsepower, edge devices, and a serious network/storage layer. One machine type cannot teach all of AI.

LAYER	WHAT IT TEACHES	GOOD FIT	WHERE IT LANDS
Student seats	AI-assisted coding, design, data work, notebooks, app builds, presentations.	Framework Desktop AI Max, Mac Studio, or GPU PCs.	Every budget.
Local AI node	Local LLM inference, privacy, quantization, model serving, latency.	DGX Spark-class 128GB unified memory systems or equivalent.	\$20K and up.
Shared GPU	Fine-tuning demos, vision, evaluation, media generation, batch inference.	RTX 5090 / RTX PRO class workstation or server.	\$100K and up.
Edge AI	Robotics, cameras, sensors, real-time inference, physical AI.	Jetson-class kits, webcams, depth cameras, small robotics.	\$50K and up.
Storage/network	Dataset governance, shared models, backup, MLOps habits, access control.	NAS, managed switches, 10GbE/25GbE where appropriate.	Every budget.



Do not sell this as hardware. Sell it as capability students can prove.

The outputs are demos, portfolios, internships, research artifacts, grant credibility, and institutional momentum.

Curriculum turns the room into outcomes.

The same spine can support a course, certificate, summer bridge, faculty training track, or CS club studio.

AI Fluency

Prompting, source discipline, academic integrity, and multimodal workflows.

Build Week

GitHub, Vercel, databases, auth, and AI-assisted app development.

Local Models

Embeddings, RAG, quantization, local inference, and model selection.

Agents

Tool calling, workflow design, approval gates, memory, and scheduled automations.

Security

Prompt injection, data leakage, access control, and red-team exercises.

Media

Voice, vision, image, video, accessibility, and presentation design.

Data

Cleaning, dashboards, notebooks, uncertainty, and decision support.

Capstone

A campus or community problem solved with a public demo and technical brief.

90 days from commitment to demo day.

The plan moves quickly, but it respects the real blockers: procurement, IT, room readiness, lab ownership, and sustainment.

WEEKS 1-2

Discovery and room plan

Confirm funding window, room, electrical/HVAC, network constraints, procurement path, and faculty/staff owner.

WEEKS 3-6

Procurement and installation

Finalize SKUs, lock quotes, order hardware, configure storage/networking, and install the local model stack.

WEEKS 7-10

Pilot cohort

Run students through app build, local model demo, governance/evaluation exercise, and presentation rehearsal.

WEEKS 11-12

Demo day and next ask

Present student work to administrators, donors, and Huntsville employers; use outcomes to justify the next funding tier.

Quality control before sending.

The fused version keeps the energy and design ambition while avoiding claims that could weaken credibility if challenged.

Kept

Kimi's executive structure, budget ladder, roadmap, risk/sustainment framing, and premium institutional tone.

Fused

Prior dark web visuals, stronger student-outcome language, local AI architecture, governance, and meeting-ready narrative.

Softened

Unverified grant-portal language, volatile exact pricing claims, and any claim that should be quote-verified before procurement.

Final procurement numbers should be treated as planning ranges until Oakwood confirms the funding source, room, vendor quotes, institutional discounts, IT constraints, and procurement rules.



Oakwood has the moment. The lab makes it actionable.

Students who learn this now graduate with a different ceiling. The goal is not AI hype. The goal is capable, ethical builders who can serve their community and compete in the market.

Prepared by Island Development Crew. Final strategic package for discussion with Oakwood University Mathematics & Computer Sciences leadership. PDF, PowerPoint, and full bundle are included for administrative circulation.